

ActInf Textbook Group ~ Cohort 1 ~ Meeting 5 (Chapter 2 pt. 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 5 (Chapter 2 pt. 2) | 1:00:12

hello everyone

it is june 2nd 2022

and we're in week five of textbook group

cohort one

we're having our second discussion on

chapter two

and we will uh get right to the

questions

first just wanted to highlight the math

learning group especially for those who

are

not regularly attending the meetings yet

the meetings are on 19 utc on wednesdays

but just with everything

don't let the synchronous time dissuade

you we can set up other synchronous

times if people want to co-work

and also the resources and the

modifications are all asynchronous

so we've been doing several things in

the

math learning group

first we've been sharing important

resources and categorizing them so if

somebody wants to

look at a video on something or find a

textbook on something or if you come

across something useful just add it here

and then if you feel like

leveraging the resources that you or

other people have added then you can

categorize it in the table and just add

a few pieces of information on it

we have a notation table that we'll be

building especially as we get into the

continuous time active inference

we have some questions

less so questions about the textbook and

more so just cool math questions that

people are raising as they're discussing

the material

and then we're building a math oriented

overview

so we've gotten up to equation 2.5

in chapter two

and this is just sort of like point by

point like what are the formalisms doing

where are the figures coming into play

how are some things related

and

any and all note taking can be added you

can always add it to like unsorted notes

or errata and then we can categorize it

later but this is how we develop a

shared epistemic resource that's useful

for everyone

and we're going to come to this in the

questions so we won't go into detail

right now

but

equation 2.5 is the variational free

energy 2.6 is about expected free energy

the g

and

this is a key equation instead of

equations so yesterday with um

uh blue

brock jessica jacob

we

went through

and line by line we went with examples

natural language readings

questions interpretations

and then jakob added um some
derivations like how are those three
lines in 2.5 related to each other how
do we go from the energy and entropy
formulation to the complex complexity
accuracy formulation
so
this is the essence of understanding the
equations
it's an infinite learning journey for
everyone so whatever level whether you
want to be
um
catching up to what has been written
here or whether you see ways to
reorganize it and go beyond and add more
connections
every single
background and level of math familiarity
can contribute in this way but we'll
come to the more details on 2.5 in the
questions
any
just
preliminary thoughts that people would
like to share
they can raise their hand or write in
the chat at any time
is there anything that anybody wants to
share before we start going through the
questions
starting with the most upvoted ones that
we haven't addressed
but we'll just pause for a few seconds
tune into the chapter 2 regime of
attention
upvote any questions that you'd like to
see addressed

post anything in the chat

yes jose

yeah i i'd like to ask if there's a
recording of the

math sessions which you had

no no

we record stick mergingly with the
traces in the notes but and we had this
conversation like about whether it
should be recorded or etc

if there's enough desire we could do
something like have some unrecorded and
some recorded this session is obviously
recorded and rewatchable here

and we do many live streams and there is
a lot of recorded material and so we
want to also hold a space for people who
don't want to

speak on a recorded version even though
everything's recorded in universe and in
each other's minds

but um

yes if there's enough interest in people
who want to record sessions we
absolutely will do it

we also just wanted to make sure that
there was a total
open math learning space where people
wouldn't even have that cross their
horizon

but we did discuss that yesterday

thank you

yeah any other like comments
people can just raise their hand with
six in gather or right in the chat so
we're going to um

just

go to the questions

and there's still time to upload like
during the discussions and we're just
gonna uh
i upload the ones once we've discussed
them so we'll start
with the first question that we haven't
discussed yet

okay this is from page 32.

policy dependent outcomes
are not immediately available but they
can be predicted by chaining together
two components of the generative model
the first is our beliefs about
how hidden states change their time
just want to find what the second
part was

the second component is the likelihood
distribution

what really changes as a result of
policy selection

the hidden states or the outcomes
bringing a jacket doesn't necessarily
change the temperature hidden state but
instead changes my observation
perception of whether it's cold or not
can we actually change hidden states
with policy selection and what would be
an example of this

does anyone want to raise their hand and
give a thought on
this question

what really changes as a result of
policy selection the hidden states are
the outcomes

while people are raising their hand i'll
give one possible thought
the answer is both slash it depends so
bringing a jacket does change the

temperature around you it's not just
changing your perception of temperature
so one example of changing a hidden
state with temperature

um hidden state of temperature with
policy selection would be like changing
the setting on the thermostat
that is intervening into the causal
process the generative process that's
giving rise to temperature which is
giving rise to your perceptions of
temperature

and so assuming that your temperature
perception is accurate you are able to
change both the temperature in the room
or the temperature around you with the
jacket

and
thus because temperatures are giving
rise to observations of temperature
also you end up changing both blue
so this was my question and i think um
carl kind of possibly started to answer
it on the live stream yesterday i need
to like think about it more and unpack
it a little bit more but like we don't
actually change

we change like the temperature around
ourselves but like we're not going to
change the thermometer reading by
bringing a jacket right so

there's like the

yeah

but anyway what carl said and what might
be beneficial for others to hear is that
um when you think about
what's changing it's like really when
you think about a hidden state there's a

causal chain underlying that hidden state and the same thing like when you think about an observation there's a causal chain underlying that observation so like no i'm not going to change what my eyes tell me the temperature is but i'm going to change what my body tells me the temperature is and so it depends because it's a multi-sensory observation right so which sense is going to change not my visual reading on the thermometer but my physical perception of gold

also yesterday on the live stream thomas parr and carl first and joined so that was really awesome and a great one to watch 45.1 and we can continue on this um ali and then mike uh yeah i think that's

what meant by the word hidden in the statement because if

it was

transparency observation to any kind of perception or observation

i i don't think it would it would be called a hidden state

i believe that the reason it's called this is it's called as such is because it's opaque to any kind of perception or observation

so it's kind of

inherent to the universe and uh probably

uh not amenable to any

uh

any change or action

thanks

mike

yeah i'm not sure if i'm thinking about
it the right way but i intuitively i was
thinking that policy selection would
uh update the expected conditions or the
expected state so maybe
adjusting the priors in a bayesian
sense

so we'll come to this later but the
answer based upon the partially
observable markov decision process
discrete and continuous time versions
policies π
intervene in between hidden states and
how they change through time
different hidden states
should they be giving rise to different
observations
will result in different observations
so changing the thermometer policy
selection is intervening in the causal
process of temperature changing through
time in the room hidden state
and if the temperature in the room is
associated through the A matrix with
your observations of temperature then
you will be changing your observations
and then just like mike said that is
reflected

based upon the expectations that the
individual has about the observations of
temperature so
they're both changing because the hidden
states and the observations are linked
that's what it means to have a
generative model

um

or for there to be a partially

observable model with hidden states and observables

hierarchical models a given state could be in emission from a higher state and emitting a lower state and that's kind of like how the current moment is like a consequence of the past but it's a cause of the future so this is related to bayesian modeling but using this architecture

we can say that policy intervenes only in between hidden states we don't directly intervene in how observations appear

any other thoughts or questions on this question

okay

all right

in figure 2.2

figure 2.2

they write

x^* and x so that's going to be generative process and generative model

latent variables hidden states do not necessarily live in the same space of measurement

it might be the

case that hidden states in the external world take on values that lie outside the space of explanations available to the brain

conversely it might be that the brain's explanations include variables that do not exist in the outside world

for example the former could be

5-dimensional and the latter

two-dimensional or one could be

continuous and the other could be

categorical

so

what are examples people can think of

where the dimensionality

of the generative model

the cognitive entity

is the same as the generative process

the niche process giving rise to

observations passing them to the

generative model

that's one question

second question how does this approach

or framing

speak

to

the map territory debate

hashtag instrumentalism and realism

what papers or live streams best

characterize how active inference models

the action cognition perception loop

and what other possibilities

or models are there for the action

cognition perception loop

what would be the different strengths

and limitations of these different

framings and partitionings

thanks um joe and then blue and then

anyone else who raises their hand

so i was thinking i mean a nice case

where you could say

dimensionality is pretty obvious is uh

board games about trying to anticipate

what my opponent is going to play in a

board game

uh

you know they have a limited number of

possible moves and

i'm assuming i'm playing the same game

i'm expecting them to make some moves on
the same board so
to me that sounds like the
dimensionality of the thing
uh because you're in this like limited
world you know i don't have to know what
they're thinking i don't know if you
know what they had for breakfast i don't
have to know
a lot of stuff i'm just thinking and i
might be wrong
but moving
great thanks for that
um

so with the same dimensionality it's
like if we're playing the same game
then um we're playing a game where we're
tracking a little cursor on the screen
the movement space within that is just
the x and the y value or just two
dimensions and so
um

the observation is like
the location in two dimensions
and then
that is um the same dimensionality that
inference is occurring on
but but it's certainly i'm thinking just
to add a little bit more of my comment
if i wanted to enrich it and say oh but
you know i'm starting to think
strategically and anticipating their
next move or anticipating maybe they're
bluffing or maybe i'm bluffing and i
want to see if they know that i'm
bluffing we add more dimensionality and
i guess in principle
again you could still have that

dimensionality

in in in somehow in parity with parts of
the hidden

states you know you should i mean that
i'm thinking oh they think they know i'm
bluffing or

um

i think they're bluffing you're talking
again about the same thing you don't
have to know what they had for breakfast
you're just talking about this extra
information which isn't in the board
itself but that gets a bit more
complicated

yes

thanks

so um

that is good answer other ones that
people have written um
thermostat

is having the same dimensionality like
it's a temperature value it's a
dimensionality of one

and so then the person's
generative model

is also on this one-dimensional
temperature axis

and then they mentioned that it could be
like categorical um versus continuous
without going too many rabbit holes deep
we could just say temperature in the
room is a continuous variable it's like
taking on a number it could be a decimal
point number between

any number

zero to infinity

um

the person's model

might be the same dimensionality like
one

but then they might be trying to make a
categorical estimate is this livable or
unlivable is it hot neutral or cold
that'd be like a three-state categorical
model

so the generative process
doesn't have to have the same
dimensionality or continuous versus
discreteness of the generative
model

second answer for same dimensionality
controls for a car have two degrees
physical degrees of freedom
the acceleration the speed whether you
can hit the gas or the brake and the
direction of the front wheels controlled
by the steering wheel

so there's like a speed dimension
and then there's a turning angle
dimension

this is the model that people use when
controlling a car at some level of
abstraction

and then the acceleration dimension
can be
acted upon

through policy selections
like stepping on the gas and the brakes
and then it's like well those are two
categorically different affordances and
then there's

continuous variables inside of that
categorical difference and we'll talk
about like hybrid control models with
both

categorical and continuous aspects

in play

and in this case like the speed of the
car in the speed axis is the hidden
state

the observable would be the speedometer

so that again the gas and the brake are
policy selections that modify let's say
we're thinking about a train no turning
policy selection

is influencing

how speed changes through time

and if you have a speedometer that's
accurate

even if it's noisy it's giving rise to
observations reflecting those changes in
speed

but gas and break do not change
the speedometer directly they're
changing a hidden cause which is the
speed

different dimensionality

rain comes from clouds simple mental
models that the darker the clouds the
more they will be

that's a two-dimensional generative
model

so

how much rain

how dark the clouds

and this also is going to speak to this
map territory

some could say well there's the size of
the cloud

and there's also the humidity so that's
like maps with increasing amounts of
variables

so no one's denying the territory of the
actual chaos of the cloud

it's just a question of how much detail
and what data we're actually treating as
observables
and
what sophistication is being taken with
unobserved variables with hidden states
which are modeled
in the computer or on paper but they're
not directly observed
then the mental model can be decomposed
even more
and then
add infant item
on the reality side
the actual
generative process
for weather is
including all the butterflies
so that's
a cognitive model
that is two-dimensional in this case how
much rain is expected and how dark is
the cloud
that one could make increasingly nuanced
cognitive models
but
whatever cognitive model is being
proposed
the generative process is something
that's like totally different cognitive
model generative model map
generative process niche territory
plants do better when they have water
and fertilizer
generative mental model leaf droopiness
is a sign of moisture but leaf size and
color are signs for nutrients
so one could imagine like a causal graph

where there's like moisture level is
unobserved
but it is emitting leaf droopiness as a
state
we're observing leaf drappiness and
we're using that to infer moisture
whereas nutrient is an unobserved state
to the visual gardener
and then they're using two dimensions
leaf size and color so this is a
three-dimensional cognitive model leaf
droopiness leaf size and color are
observables
and then there's um two
unobserved cognitive states nutrients
and moisture
but whatever map gets constructed that's
not the plant now the gardener might
then want to take an informative
experiment
being an optimal bayesian gardener and
then measure something take something
that was
an unobserved in the initial model and
then measure
using lidar like using some sort of
sensor to understand and then that would
make that data point
an outcome
related
through some a matrix which we're going
to get to later
of the hidden state being like the true
levels of copper and then if we have an
accurate test we can start using the
outcomes of the observed levels of
copper and start
flushing out our model that way joe

i chucked in a comment uh below i hope
it's showing up uh
just the
other part of the question was asking
about the map territory
uh debate and and here's from the person
who kind of came up with that uh
that dichotomy in the way we we usually
talk about he says that the map and
territory should have a similar
structure and i think that's really
interesting to think about you know if
i'm making again moves in a board game
i might
you know make a reduction of their
mental state they may not maybe i'm
playing against the computer for example
so i don't know that they have a mental
state or maybe i'm playing against um
a person i happen to know as the
beginner so i know they're likely to
make naive mistakes or whatever
something like that um but i was just
thinking like with regard to the leaf
droopiness thing you could have all
kinds of really
badly badly structured uh models of
plant health
um you know like
uh maybe relate to how recently you
watered it or something that sounds good
in general but if it's a cactus
maybe that's maybe that's like a
defeater to your to your system so um
the point being like it seemed
interesting that your map should maybe
if it doesn't have exactly the same
dimensionality should still be a nice

reduction of the of the thing you're
trying to model whatever that means
thanks
one
aspect on this is
evolution natural selection dissipation
in our world sweeping off the table
entities that are failing to at least be
adequate
so
in a dissipative situation that we're in
the map has to be at least good enough
otherwise the entity will fail to exist
so that sort of closes the loop
and
is like we're seeing the persistent
entities that are acting good enough to
navigate
um and it's just interesting like how
many
from the allegory of the cave to the the
blind people and the elephant and
realism instrumentalism
and we've had many live streams
uh on this
especially check out number 14 on mel
andrews paper the math is not the
territory
so this one sort of explicitly jumps in
right there and tackles this issue with
the fep front and center so this is a
great series of live streams um the dot
zeros
are background in context where we just
go over the paper in a small group like
one person or two or three people
and then the dot one of the dot two we
usually have more of an open

participatory discussion and the authors
sometimes to usually join so everybody's
welcome to participate
in contributing to dot zeros
as well as enjoying the dot one of the
dot two but just so you know when you're
looking
in the live stream table
um the dot zero is a good one to watch
first
because it has like the background and
overview on the paper but then like
yesterday in
45.1 we had fristen and par join
to discuss just so that people can
navigate the live streams a little
better
um so we can continue
um asynchronously adding more
uh
thoughts on like map territory and what
papers and live streams and what are the
possibilities for the action perception
loop but this is these are big
fun
areas but we're going to continue with
the questions
on the notion of surprise
is the agent's
perception not only affected by the
influence
of not only its environment but also an
agent's peers
so perhaps three state is in how is an
agent's perception
affected by its direct perception and
its assessment of peers
so the agent's peers are updating their

beliefs in a collaborative fashion

what would be the extent

of the update to an agent's generative

model of perception if the agent

witnesses the annihilation of one of its

peers for example

something like this

okay

wildebeest from bird box not gonna open

any video links but perhaps we'll

observe some annihilation like some

animal attacks one of the wild beasts on

the right but then like the other one

learns

so

this is a great video definitely good to

watch

um

so far in the textbook

and indeed for the textbook it's

focusing on

just like we saw in figure 2.2 it's like

one entity

in the niche

the niche can consist of others like me

other entities that i expect are similar

to myself in terms of like the

coarse-grained cognitive architecture

they have or their preferences their

affordances they're able to do similar

things they want similar things they

have a similar history

to some extent

so

um

that is sometimes called thinking

through other minds to me

and just to give one thought and then

anyone can raise their hand
um
if we were to accurately observe
an agent
another peer
taking some action and then failing to
exist we're driving on the highway and
then we observe somebody taking a policy
to go off the highway and then we see
them fail to exist
we could use that information
to update
our
beliefs
about the consequences of the action to
drive off the highway
so
i believe it would be possible
to
implement every variation of
direct perception direct perception of
peers thinking through other minds
theory of mind
learning by example
imitation contrarianism
every phenomena could be modeled as just
has to be
whatever the specific
um
model is actually about ali
i have a related question about
2.5 and 2.6 uh well uh correct me if i'm
wrong but uh
i understand that these equations uh
describe a kind of
uh
describe the behavior of a kind of
rational behavior the kind of behavior

of a rational agent
uh where does irrationality comes into
this um
or
let me put it this way
can irration can irrationality be a
parameter uh
in order to qualify these two equations
2.5 and 2.6 or seen
from
the other side
can we use these two equations uh to
measure the extent of rationality
as a
measurement of rationality
good question
here's one quote and we'll come to
equation 2.5 and 6 as well because those
are important
distinguishing the generative model and
the generative process is really
important
that's the difference between the entity
and the niche it's important to
distinguish them to contextualize
psychological claims about the
optimality of inference
to the extent these claims are valid on
a bayesian view is always contingent on
the organism's resource
so
don't want to step out too far but
rationality and irrationality
it's rational to believe in x it is
irrational to believe in x is always
subjective from that speaker's
perspective
given the priors and the update rules

all there is is rationality which is to
say this is just the bayesian updating
process so there are maladaptive priors
ones that

trend towards dissolution of the entity

but there aren't irrational priors

there's just is what it is and how it

updates and then that could be

inadequate

um it could be any number of other

things and these are like getting into

relatively complex cognitive phenomena

um where there could be like

multiple layered models

but um

i would say

rationality in the base process is like

the cognition is bayes optimal or

modeled with a bayesian process

and then sometimes the interaction of

the generative model and the generative

process as appropriately defined

specifically

is going to result in what it results in

so if

but let's we will come to 2.5 and 2.6

but okay here suffice to say

yes

social learning and collective behavior

and emergent

uh

outcomes at the group level is something

that's important but we haven't

addressed it so far in the textbook

and multi-asian models are not really

addressed in the textbook that much

because there's so much to understand

about the kernel of the perception

cognition action loop

that

even though it's so important for real
systems

it's not brought up

okay

wanting to go to the

um equation

ah the dark room problem for many years

many papers firing back and forth

but let's go to the um equations for our

let me check the chat here yes

okay

variational free energy is minimized

through two different possible

approaches

minimizing divergence

of q and p changing mind

and maximizing evidence

taking action changing perception this

is done based upon prior and present
information

equation 2.5

variational free energy

to plan

best actions generative models and

internal policies produce simulated

outcomes that can be used to estimate

expected free energy as the average of
the log probability of outcomes

equation

2.6 expected free energy

and there's other variants

in the context of planning planning as

inference

equation 2.6 provides a view of efe that

establishes consistency in measurement

units of exploration exploitation

the relative balance between these terms
determines whether behavior is
predominantly explorative or
exploitative

34.

in other words dissolving the classic
explore exploit dilemma in behavioral
psychology

what causes this balance to change i.e
what controls this balance very
important and interesting question

perhaps we can come to equation 2.5
first and then we can come to equation
2.6

so

um where we're going

is the future

the future has several sources of in
principle and in practice uncertainty
observations in the future have not
happened

actions in the future have not happened
and the causes

of future outcomes and hidden states
actions as well as the generative
processes endogenous changes haven't
occurred so there's several like

fundamental and in practice

limitations of

the kind of precision that you can get
on the future

that's planning as inference it's
prospective

in

contrast very and that's expected free
energy because it's about the free
energy of an expected future
variational free energy f and expected

free energy is G

variational free energy F is in equation

2.5

variational free energy as the question

says

is based upon present and prior

information

so variational free energy is like

now casting

and

verging into something kind of like

memory as inference we're integrating

the past

in the sense of our priors

and the present incoming data point y

and it's like a snapshot

evaluation of optimal perception

in the purely sensory case

and if we include action within

variational free energy it's taking

one-step optimal actions

but those are not necessarily like the

ones that that's not planning that's

just like snapshot decision making

in contrast 2.6

will explicitly consider

um

policies it is a function of policy π

and so it is including planning as

inference but first let's look at 2.5

um

jakob or anyone else

would you like to just describe

one pass

what is equation 2.5 showing

um

i guess if you mean uh like describing

say the first line or the lines in a bit

more detail

essentially it's the

um

so the

the energy term is just the expectation

of

the log of the joint probability of the

data and

the observation so i think

um

i don't understand quite why they

changed

the notation from appendix b

but

why would uh essentially be the s i

think and x would be the o

yes um anyone could be taking more notes

on this but y is the observable data

and x is the hidden state

in the pomdp that we're going to be

getting to later

x

out there the hidden state is s and o

are observations so this is using more

of a like regression

familiar like framework and variable

notation

but

this could also be written $\ln p$ of

o comma s observations and states that's

the joint distribution joint

distribution with a comma conditional

distributions with a vertical pipe

yeah and so we

also taking the the entropy term is only

based on

the beliefs of our hidden states so

for your

how
um people were saying before that the
hidden states are unobservable
the only thing that we can say about the
hidden states are the belief
cue
about the hidden states
i don't think that the
hidden states x can ever really appear
um on their own
because we always need to have some kind
of belief about them and
this was also related to a discussion we
were having yesterday
um that's interesting that they
uh used to
kind of switch between the notation of
entropy
with h
um
as the func as the functional
uh and the
and just the pure expectation uh
with the e
because the actual definition of entropy
is the expectation of the negative log
of
whatever is in
um
[Music]
is um is the variable which in this case
is q of x
uh and that's actually how we get to the
other
to the other two expressions with
complexity and accuracy and divergence
and evidence and there's also a link
i think might be wrong with uh

that to theoretical physics with
um where
free energy is essentially
the kind of cognitive statistical
equivalent of the lagrangian
which is defined as kinetic energy minus
potential energy so we were discussing
uh yesterday
what that means in terms of uh the
energy and entropy in this case whether
the energy is
perhaps
kind of the immediate um
[Music]
cognitive kinetic energy
and then the entropy
is like the potential energy or vice
versa
[Music]
might be going off on a tangent at this
point
awesome let me just add in a few other
notes
um the variables
 f is a function
it's the variational free energy
function
the arguments that it is taking in
our q
that's the variational distribution that
we control that's what makes it a
variational free energy
 q is the one that we control
 p is the actual probability distribution
out there
 y is the data point
so we have on the left side of the
equation

variational free energy is a function of q and y

on the right side we have three equal signs and these are three values

three ways to phrase this one function in the bottom here

Jacob has shown how this energy minus entropy

can result in the second line complexity minus accuracy

and in the third line divergence minus evidence

so that was like one really

interesting thing is it's not like this

is how it is and this is a

transformation and then this is the transformation it

even in a sense that is also true it is

also probably helpful to think about this

energy minus entropy formalism for what it is

which is the closest to a physics framing

of free energy

and then think about these two more statistical

ontology

ways of framing it complexity minus accuracy which is like very commonly

brought up in the context of model

fitting and divergence minus evidence also

um

and yeah great so blue if that sounds like a very helpful norm

in the description

we'll try to have like a description of
the equation
the first row says this this this this
this
people can type it however they want and
then of course we can replace
terms that are in the ontology
with
the special at sign
and that will facilitate
finding equations that mention those
terms
and it will facilitate translating these
equations
and
also the descriptions into different
languages computer languages and human
languages
so then we can work on it in this
ontological space
and then make sure that it's accessible
to different languages for example
um
then we thought about this example of
like
uh a ball
and its location in a bowl
so
there's a true location of where it is
in the bowl
x
and then there's an observation
of where we're observing the ball to be
so
and this also speaks to the quantum to
classical handoff
if the bowl is a swimming pool and the
ball is a bowling ball

after enough time it's going to be at
the bottom and not moving
if
the bowl is like a molecular well
and the ball is like
an electron or like one atom
then repeated measurement is going to be
entirely different and as the
ball and the ball become more massive
it approaches like a classical limit
little hints at the continuity between
classical
statistical
and quantum and thermo physics um but we
thought about this ball being in the
bowl
so
um
we looked at two different cases
which is when the ball is being observed
in the bottom of the bowl
it's like
given the generative model of the shape
of the bowl and gravity
um which is like a potential energy
function
observing the ball at the center of the
bowl is like strictly the most likely
thing that could have happened
and then observing it up on a side is
like a less likely observation
um eric and then blue
i
see uh yeah um i so it you mentioned
that uh f is a function and and as they
put it it's a function of a function
which is a functional so the way i think
about this is that

the function that it's a function of
is the q so the q is a distribution over
your beliefs
so i think of that as being a function
that you're trying to optimize you're
trying to say well how do i figure how
do i adjust my internal
belief state the q
and that's what i'm optimizing
in this equation and the functional says
that's what i'm trying to optimize is
that function and distribution and then
it breaks down into the two parts which
is
your your generative model p which says
well
um
 p says for any
there's the joint probability of x
 x is the the internal state
um
variable that you have a distribution
over that's q
and so how do your observables and your
internal states go together and then you
have the entropy term which is well you
want to have
maximum uncertainty so that if you're
not constrained by your observations you
want to be as agnostic as possible about
what the possible values are of your
internal states or make that q as flat
as possible
as flat as possible under the
constraints of the observations going
through your generative model p
so i guess that's the way i talk through
this this first line awesome thank you

eric blue and then brock

so eric that was super helpful um i've been like since yesterday trying to figure out what exactly is meant by q here

like i get like later on we're going to talk about q as representing beliefs but in like as it's given as it's laid out in chapter two like they don't actually define q at all

um

so i am like having a hard time up here

like they describe it

much like right before they get into that box

they describe q in section 2.6 as an approximate posterior

so

approximate posterior of what and like should we

denote q as the function of beliefs here even though

like there's it's like they're skipping ahead and looping back at the same time

it seems like or it feels like to me

q is the variational distribution where the form of the posterior has been chosen

so exact base

using matrix multiplication basically or sampling based approaches to approximate bayesian computation like monte carlo markov chain

those try to recapitulate the actual form of p

so if p is truly bimodal then exact base

or sampling based approaches would try

to reconstruct that distribution
in contrast the variational approach
gains tractability because you're
choosing a posterior approximate
posterior cue
of a form that you know is going to have
a tractable optimization scheme and
there's message passing algorithms
that can do it computationally so if you
choose q to be a gaussian then
depending on how you parameterize it
it's either going to like be seeking the
mode and go to the hump that's bigger
and center around that with a variance
or it'll just spread out a lot and go
over and try to cover as much
probability density as it can by
covering both humps
okay so the variational free energy
function f as a function of the function
representing the approximate posterior q
and the data y
is equal to the negative expectation of
 q given the observations x times the
natural log of the probability of the
joint distribution of the data and
observations is that the english
translation of that equation i think i
got it finally after like two days
minus close
close i would say you one one thing i
would say slightly different
it's the expectation over q not
expectation of q
we actually talked about that um a
little bit like here we had negative
expectation over q
oh and

yes

um okay

blue then yeah sorry the ontology terms like don't match the terms that are given in the book and i i put in a request in the questions here if we can append the ontology because they use like data

is the same as like

um

observations

those are different words but there's they're similar

like yeah there's a lot of of synonyms or synonymous use and so like i would hate to step on them or but like also the terms aren't

um

in the ontology at all so maybe we should update the ontology to reflect the terms

we will absolutely add terms into the supplemental or eventually into the core terms if that's required but just

leave a comment or email active inference at gmail to add terms to the ontology if there's a term that you want to tag that's not being tagged
brock

i feel like you may have indirectly answered this just now but um it's in the math discussion yesterday we were talking about specifically about the cube the distribution of beliefs um over the yeah like

if x is this uncertainty term this

um

hidden state like

what

what value could it possibly have where

is it coming from

or is that

so q is this property of distribution of

beliefs is x the belief

value and that's being generated

in the previous step

i think

we in the bowl

marble thing we

like my question was uh

or i was just pointing out i guess that

you know the hidden state is changing

however it wants um perhaps below our

threshold of observation

that's not reflected in the equation for

obvious reasons and so but but then

where you know maybe it crosses a

threshold and then we update or whatever

but but where is that

x coming from if it's a hidden state

you see what i'm saying is that kind of

an ontological problem there i'm trying

to solve for it's like it's a hidden

state how can have a value or be a

variable if it's hidden

it's a

yes eric

i'll take a stab at that

so as i as shown in the pictures there's

two hidden states

there's a hidden state of the real world

and then there's a hidden state of your

generative model

and the general model is whatever the

model is and that's however the model
was built so once you have the
parameters x
then that's what you're working with
until you change your model
so that was my
habitat

yes
and over evolutionary or developmental
times

you get swept off the table if there is
inadequacy at least

but in a simulation environment you
could

make the x and x^* to be anything but
um

the hidden state is as modeled by the
generative model of the cognitive entity
so

in the cognitive model of the entity in
the bowl example they're they're
modeling an unobservable which is the
location of the ball

and they're modeling an observable which
is the observation of location

so both of those are in the generative
model

and then the generative process would be
like there's somebody with a magnet
who's moving the ball around or
something or it could be anything
um

we're going to have five more minutes
before we close this session

so

what does anyone see in 2.4

what is

free energy

minimization doing
and what does it mean that variational
free energy is an upper bound on the
negative log evidence
just one quick qualitative read
is that this is what we would actually
like to be minimizing this is the floor
we truly want to reach would be
unsurprised
about observations
however
that
is intractable because we don't have
access to the form or the
parameterization of the true generative
process
what is a tractable approximation
approximate bayesian computation
is variational bayesian inference so
here's $f(q)$ of y
 q and y
and then this is a kl divergence
one vertical pipe means conditioned on
like x conditioned on y x given y here
probability of a hidden state being the
case given a data
or the other flipping which would be the
probability of a given data point given
a hidden state
two vertical lines is in divergence
notation
and that's the divergence between this
and that
between q of x and between p of x given
 y
so
that
reducing this divergence

which has very nice properties in terms
of its
implementation
is bringing us to lower our upper bounds
on what we really want to be minimizing
in a tractable way so that's one aspect
on free energy minimization
and then um
here is reframing the two ways that free
energy variational free energy can be
minimized
through perception
the updating of beliefs and we'll talk
more about the continuity between like
perception and learning
like if one observes the ball moving
across the visual field is that
perception or is it parameter learning
updating the location of the ball that
depends on the time scale and such
or action can be taken
so that the expected observations are
changed to maximize evidence
and then
in the last minutes
ali asked
can a hidden state be an unquantifiable
quality
like a qualia quality or like an
unquantifiable quantity
no i mean uh well
of course most qualities can be
parameterized but
well
perhaps
we can think of a hidden state as a
pure quality
that cannot be

is fundamentally unquantifiable and
can we we cannot parameterize it i mean
can we think of such a hidden state
um
brock

i was just going to say the probably the
quantifiable way to state that question
is like are there
intractable

hidden states which the answer to that
is like obviously
yes

can we get some estimate on it yes
but then is that

commensurate with the i you know and
then does that mean it's a quality and
not a quantity i don't know
and also to the earlier discussion about
like the dimensionality of the
generative model and the generative
process the unquantifiable part could be
like the baby's happiness

but then

we're able to still have an inference on
the happiness as a parameter
x the underlying state that's giving
rise to the different sounds
without

necessarily knowing like the
distributional form of happiness but we
have to have a distributional form for
the variational approximation of
happiness joe

i was just going to say i mean i think
that this

qualitative versus quantitative
distinction is interesting if you're
thinking about like a light source you

know it's a source of electrons well you could count the number of electrons but whether or not there is a light source they're a source of electrons in this framework and i'm proposing you're not counting that you're just saying there is a light source or some other dimension the thing we've been calling dimensions you know yes you can count the number of dimensions but given that they're all different they're not really relatable to each other

thanks yes very interesting um so in our final seconds we'll just point towards 2.6 which is expected free energy

g

and its prospective which is going to enable inference conditioned on action x hidden states the world conditioned on policy selection and that is what is going to allow comparison of different policies which are sequences of actions over a given time horizon so minimizing the divergence between two things minus some things reframing it as a few things so we didn't go as much into 2.6 as we did into 2.5 because 2.5 is an important precursor and simpler cousin of 2.6 but it's like super important and we especially to those who are here and those who are not here we would

really love over multiple cohorts when everybody will hopefully be able to join and rejoin as participant as facilitator um to be able to annotate all the equations with what they mean because what is there to be said about expected free energy without some understanding of what this is actually discussing what variables go in what is it doing how are some of them being related so we want to learn this and we can do it collaboratively if people just pick and choose affordances they see annotate some things and then they can highlight it and say i'm not sure about this but i just wanted to add it can somebody improve it and so on like we're all there to help each other learn and improve our epistemic niche

2.6 is expected free energy there's further discussion on expected free energy showing how it's composed of these five sections and that by like leaving some out you get different special cases that are quite important for example leaving out this you get this leaving out just one you get two three four five and so on figure two six and then there's a summary of the low road to active inference that's chapter two

so this concludes chapter two and our discussion in cohort one

in the coming two weeks we're going to be discussing chapter three the high road to active inference

it's going to take another attack

and we will go through it in the coming two weeks and be addressing people's questions but hopefully we can continue answering and addressing just wherever we see fit no one's gonna do it all but we do need everybody to do some hopefully

we're going to now close recording we'll then take a one minute break and then in this room we're going to continue with dot tools

if you want to continue discussing the textbook or anything else then head up to a room

up and to the left but in this room in one minute we're going to continue with dot tools so thanks everybody see you soon